Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/08/2012
PI Name:	Oman, Charles M. Ph.D.		
Project Title:	Validation of Assessment Tests and Countermeasures for Detecting and Mitigating Changes in Cognitive Function During Robotics Operations		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	07/01/2009	**End Date:**	07/31/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2013 per NSBRI (Ed., 7/2/13) NOTE: change in Team to Sensorimotor Adaptation Team, from Neurobehavioral and Psychosocial Factors Team, per NSBRI report (Ed., 8/8/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Czeisler, Charles (Brigham and Women's Hospital) Liu, Andrew (Massachusetts Institute of Technology) Lockley, Steven (Brigham and Women's Hospital) Barger, Laura (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-NBPF02001		
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Performance Goal Text:			

Task Description:	The goals of this four-year NASA/university project are: (1) characterize the changes in cognitive function during robotic operations that affect performance, (2) validate proxy cognitive and drowsiness assessment tests as predictors of performance changes in a complex operational task, and (3) test the efficacy of fatigue countermeasures (e.g., blue enriched white light, caffeine) to improve cognition during robotic operations. To date, the complex robotics operations aboard the Shuttle and ISS have been successful, but not entirely without incident. Fatigue is common because of chronic sleep deprivation and circadian misalignment of the sleep-wake cycle due to unusual sleep schedules or abrupt "slam shifts" in crew work schedule, and the long duration of robotic operations (up to 6-8 hours). The impairment of cognitive abilities has been identified as a factor leading to errors in performance. In nominal operations, pauses, unnecessary movements and performance inefficiency may reflect cognitive impairment. Preventative countermeasures are necessary to reduce this risk and improve crew safety for current and future robotic operations. This project is a collaboration between the MIT Man Vehicle Laboratory (MVL), and the Division of Sleep Medicine at Brigham and Women's Hospital (BWH), with the assistance of collaborators at Jefferson Medical College, NASA Kennedy Space Center, and the NASA Johnson Space Center Astronaut Office and Mechanical and Robotics Systems Group. Subjects are trained to perform simulated robotics tasks under realistic astronaut schedules. Using a within-subjects design, our experiments investigate how the proposed fatigue countermeasures affect physiologic, cognitive and task performance metrics, thereby enabling their relationship to be understood. If this correlation can be established, it will provide evidence that these metrics could be valuable as indicators of fitness-for-duty. Subjects are initially consented, screened for robotics aptitude and medical factors, and then follow a normal sleep schedule (8 hr/night) for two to three weeks and receive further robotics training. Robotics task include a variety of simulated ISS fly-to/grapple, autosequence, and track-and-capture tasks. Subjects then shift to a restricted sleep schedule 6 hr/night for one week before admission to the sleep clinic for the inpatient phase of the study, and undergo a refresher training session. The inpatient study design consists of four three-day-long experiment blocks. In each block, subjects take a 3 hour nap late on the first day. On the second day they are awoken 9 hours early - a "slam shift" of their sleep schedule - and their performance on robotics, concurrent mental workload (imbedded visual response time task), cognitive (Number-Letter, Digit Symbol Substitution), and psychomotor (10 min PVT) tasks is evaluated during a 6 hour robotics session. They will have been awake for 18 hours by the end of the robotics session. Subjects are allowed 8 hours of recovery sleep on the third day, on the preadmission schedule. Physiological measures (plasma melatonin and cortisol, EEG, Optalert eyelid opening speed) and subjective sleepiness (KSS) and alertness measures are taken throughout. The first 3 day block is used as a control: subjects are tested using 90 lux white light. In subsequent blocks, subjects are tested under three conditions with order randomized across subjects: a) 90 lux blue-enriched white light, b) 90 lux white light and hourly caffeine (0.3 mg/kg), and c) 90 lux blue-enriched light and caffeine. The blue-enriched light stimulus is provided by a Solid State Lighting Module developed for ISS. So far, 18 of 24 planned subjects have completed the inpatient study or are in process, and initial data analysis is underway. As described below, this year MIT also completed two pilot studies: The first confirmed that subject spatial ability scores of our inpatient experiment subjects predicted performance during initial robotics screening by MIT trainers blind to their spatial ability scores. The second, conducted on MIT student volunteers, demonstrated that imbedded visual secondary tasks are sensitive to 18 vs 4 hours awake, even though subjects were apparently able to maintain their performance on the relatively challenging primary robotics task while sleepy. Findings suggest that a decrement in spare attention may be an early sign of drowsiness when performing demanding robotics tasks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An effective, safe, well-tolerated, non-invasive countermeasure for circadian- and fatigue-related deficits in cognition is required for use during in space environments to enhance the safety of crew members. Light exposure has the potential to fulfill this role. Although monochromatic blue light exposure at night has been shown to be most effective at shifting the circadian pacemaker, and improving alertness and performance, these results have not been tested for broadband blue-enriched white light or for light exposure during the biological day. Before light therapy as fatigue countermeasure is operationalized, further research is required to fully understand the effect on complex performance, such as robotics performance. Similarly, while caffeine use is widespread, including on the ISS, uncontrolled use of caffeine may not be optimally timed or of the correct dose to alleviate sleepiness on duty and may interfere with subsequent sleep, thereby increasing fatigue the next day. The current study will investigate the effects of light and caffeine as fatigue countermeasures in a controlled environment with the long-term view to developing specialized countermeasure schedules to maximize alertness and performance of space crew in an environment where even small fatigue-related errors could have catastrophic consequences. These results will also have widespread application on Earth. Approximately 18,000 Air Force pilots and flight crew are exposed to conflicting circadian signals and resulting fatigue from long flights. Tens of thousands of troops are subjected to rapid deployment world-wide and are often required to maintain alertness while suffering from jet lag. Additionally, nearly 180,000 troops are required to be vigilant for more than 24 hours at a time, and thousands of Navy and Coast Guard personnel work for extended periods under circadian misalignment. Over 200,000 commercial airline pilots, flight crews and air traffic controllers are also subjected to extended shifts and circadian misalignment. The NTSB implicated hours of duty and fatigue issues in the 2009 Colgan Air commuter airline crash in Buffalo, NY, and this year the FAA has overhauled rules for air traffic controllers after several highly publicized issues where controllers apparently fell asleep. More than 230,000 ER doctors, staff and medical residents work extended shifts for 30 hours or more. Approximately 1.9 million public safety personnel (police, corrections officers, firefighters, etc.) work extended shifts and are exposed to emergencies which require vigilance for long periods. In total, more than 3 million Americans would directly benefit from the findings of this research. In addition, applications may be possible in standard occupational settings to improve general day-time alertness; in educational settings such as in training sessions and conferences, colleges, and schools where enhanced alertness would assist learning and memory; as a fatigue countermeasure for sleepy car and truck drivers in order to provide them with a short bursts of alertness that will allow them to find a safe place to stop driving and take a break. These wider applications have the potential to reach tens of millions of customers and indeed anyone who could benefit from improved alertness and performance.

Task Progress:	As we complete the 3rd year of the project, we have now completed testing of 15 subjects with 3 more in process. Interim data analysis is underway, and we plan to conclude testing by mid-fall and initiate final data analysis of data from 20-24 subjects. Admission of inpatient subjects has been complicated since in addition to the usual screening for medical/psychological reasons, we found that some subjects had difficulty learning the robotics task. We now evaluate their performance during an early training session, and have released 28% of subjects. Earlier (Liu, et al., 2012) we showed that subject spatial ability test scores were moderately predictive of astronaut performance during NASA robotics training. Our experience screening fatigue study subjects confirms this: All our study trainees took the same battery of four spatial ability tests administered to our astronauts. We standardized our trainees' scores on each test using our 319 subject database, and averaged and ranked the results. Our screeners were blind to trainee spatial ability scores. A logistic regression on average rank yielded a predictive relationship, with an ROC-AUC metric (Fan, 2006) of 0.96. This value is higher than obtained by Liu et al., probably because the spatial abilities of our trainee population are more heterogeneous. The result did demonstrate the advantage of using a composite of four spatial ability tests in predicting performance and will be included in the Master's Thesis of R. Galvan. This year we also completed a second pilot study to understand how side task response time covaries with robotics performance as a function of time awake, and whether the task is simple detection or two choice. 13 naïve MIT subjects performed two types of robotics tasks. Response time (RT) to either a simple or a two choice visual secondary task was measured. Subjects completed two 2-hour robotics sessions, one at midday after approximately 4 hours awake, and one at night after 18 hours awake. Comparing 18 hours awake versus 4, KSS scores increased by at least 2 points. Subjects apparently rallied to maintain robotics primary task performance through night sessions, but secondary task inverse RTs showed significant changes in spare attention, with useful Hedges' g (0.35 to 0.74) effect sizes. We concluded that our visual secondary tasks were sensitive to changes in primary task mental workload and sleepiness, even when primary task performance was not. A secondary task does not require separate testing like the PVT. Increased RT can be interpreted as a decrease in spare attentional reserve provided the primary task does not change and training effects have disappeared. Secondary task workload measures deserve further investigation for use in detecting sleepiness. Detailed results are included in the Master's thesis of C. Lowenthal (2012), are being presented at the 2012 Boston sleep meeting, and in a manuscript.
Bibliography Type:	Description: (Last Updated: 12/18/2024)
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Dissertations and Theses	Lowenthal C. "Evaluation of Sleepiness in Space Robotics Task Performance and Discussing Sleep with High School Students in a Museum." Master's Thesis Dissertation, Massachusetts Institute of Technology, Cambridge, MA, February 2012. , Feb-2012
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